EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

EMEM gives an observation a lookup identity and a content-addressed record. Agent A passes the reference. Agent B resolves the same record and checks it before continuing.

Earth data stays in the data layer. Its reference enters the reasoning layer.

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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortx-AI/emem (Apache-2.0)



1 How does Earth data enter an agent workflow?

Keep the source data upstream. Pass its reference.

Locate an observation, record what was read, and carry its content address into the next agent's reasoning.

EARTH DATA → AGENT REASONING

OBSERVE — Read the source — Optical · radar · terrain
→ **LOCATE** — Name the lookup — cell · product · time
→ **RECORD** — Canonicalise the record — value · derivation · sources
→ **HAND OFF** — Agent A passes a reference — emem:fact:<cell>:<cid>
→ **RESOLVE** — Agent B retrieves it — the exact addressed record
→ **CHECK** — Bind and re-hash — attestation · recipe · source
→ **CONTINUE** — Use the checked record — compare · reason · act

Lookup identity: where / product / time

Content address: BLAKE3 of the canonical record

CONTROLLED HANDOFF TEST: ACTED ON CORRUPTED EVIDENCE

prose	JSON	retrieved text	opaque id	checked reference
254 / 276	164 / 276	220 / 276	154 / 288	0 / 300

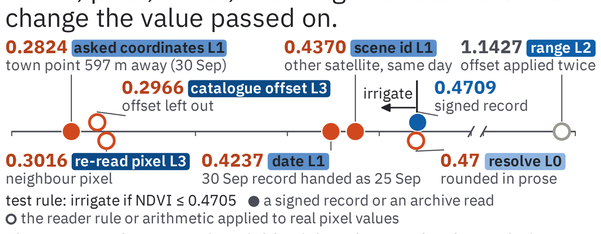
In the controlled handoff test, prose led agents to act on 254 of 276 corruptions; a reference with explicit checking instructions, 0 of 300.

Claude Haiku 4.5, Sonnet 5.5 and Opus 5.5 as B, pooled; Qwen2.5-7B apart; 2,794 scored trials on 1 Oct 2026. Pooled results above are the Claude runs; Qwen is reported separately.
Qwen2.5-7B: prose 22 of 23; checked reference 2 of 25. One run. Simulated corruptions, constructed decision threshold; model breakdown and confidence intervals in Reproduce.

2 What can change between agents?

One question, several records

Scene, pixel, offset, rounding and stale state can change the value passed on.



test rule: irrigate if NDVI ≤ 0.4705 ● a signed record or an archive read ○ the reader rule or arithmetic applied to real pixel values

The same Keylong question yields eight values under these choices. Keeping the cited record makes the differences inspectable.

3 Where did the evidence change?

A changed value needs a cause

Keep the scene, pixel, model and time attached so a change can be investigated.

$$\Delta z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \varepsilon$$

World, instrument, alignment, model and noise. This is an attribution model; the numeric split remains open work.

IMPLEMENTED · DRIFT-ANCHOR SCORE

$$u = |x - a| / (3\sigma)$$

$$s = u / (1 + u)$$

x: output · a: anchor · σ: anchor uncertainty

consistent — tension — contradicted

0 — 0.50 — 0.75 — 1

Tests pin 3σ → 0.50 and 9σ → 0.75.

Missing or invalid σ: exact agreement required.

The implemented anchor score measures disagreement, not its cause. SAT-042 exercises it in a reference harness; live device-to-anchor wiring remains next work.

Memory = observations + edges

$$M = (O^*, E^*)$$

O* stores observations; E* stores typed temporal relations. Append new records and keep earlier ones.

$$E = (subject, predicate, object, t_{from}, t_{to})$$

Relations can say supersedes or disagrees_with, with a validity interval.

Lookup: (cell, band, observation time)

$$cid(O) = BLAKE3(CBOR_{canonical}(O))$$

$$recall(M, a, b \mid t^*, \tau)$$

At address a and band b, bound observation time by t* and what memory knew by signing time τ.

Model outputs are records too

Satellite input → encoder → vector record → content address

Prithvi 1,024 values

checkpoint digest inside the record

TESSERA 128 values

product year 2024; no checkpoint digest

Archived vectors: Prithvi names a checkpoint digest; TESSERA names a product year. These encoders are retired on the deployment; stored records remain addressable.

Missing data stays explicit

An absence record carries a reason and source. Berlin's CHIRPS record says outside the product's latitude range; it supplies no rainfall value.

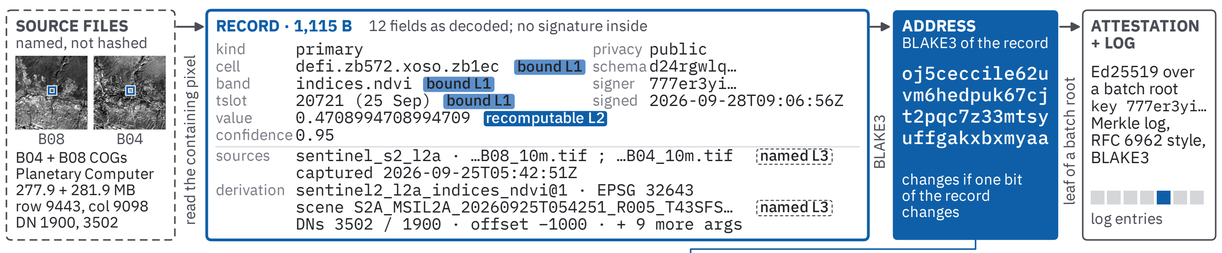
Where emem fits. STAC finds assets; openEO runs processing; PROV and C2PA carry provenance. EMEM adds the observation reference an agent can resolve and re-check.

4 What exactly is handed over?

A compact reference opens the observation record

Resolve the value, place, time, derivation and source references from the record's address.

INSPECT The 1,115 bytes behind this example.



`emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk67cjt2pqc7z33mtsyuffgakxbxmyaa`

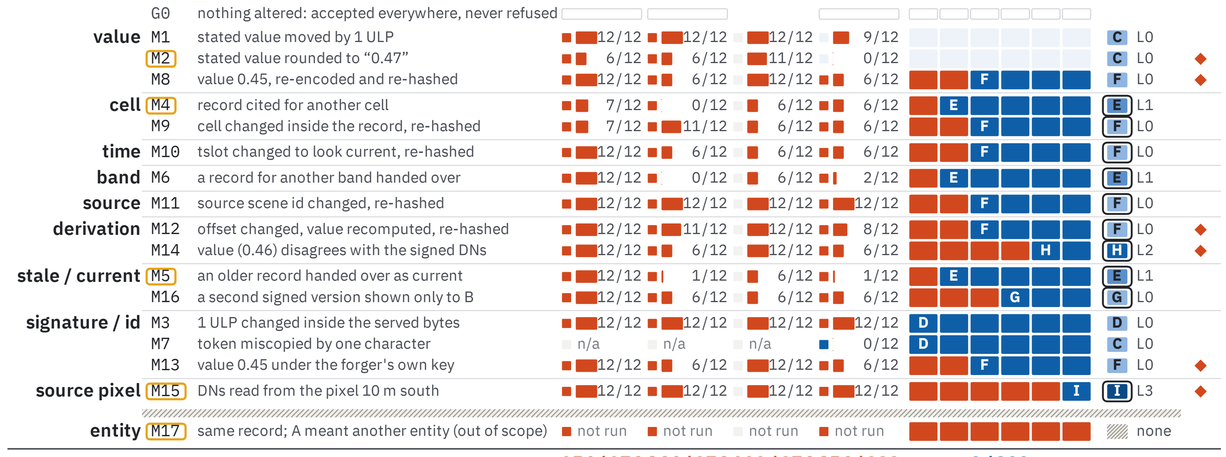
handed over: 84 characters, 46 tokens (cl100k)

BLAKE3 identifies the exact 1,115-byte record. A batch attestation covers its address. The record names the source files and read location; source checking continues by re-reading them.

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve → Re-hash → Bind → Recompute → Re-read



B acts on corrupted evidence

prose	JSON	RAG	opaque id	EMEM, full check
254 / 276	164 / 276	220 / 276	154 / 288	0 / 300
R1 15/15	R1 15/15		R1 13/16	R1 0/16

C resolve L0 · D hash L0 · E binding L1 · F signature L0 · G log L0 · H recompute L2 · I re-read L3 · D to I are cumulative: each adds one check

Bars: agents' false acceptance, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 items, 1 Oct 2026. Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Each step answers a different question. The matrix measures which changes become visible at each depth; signatures and log checks establish attestation and publication history.

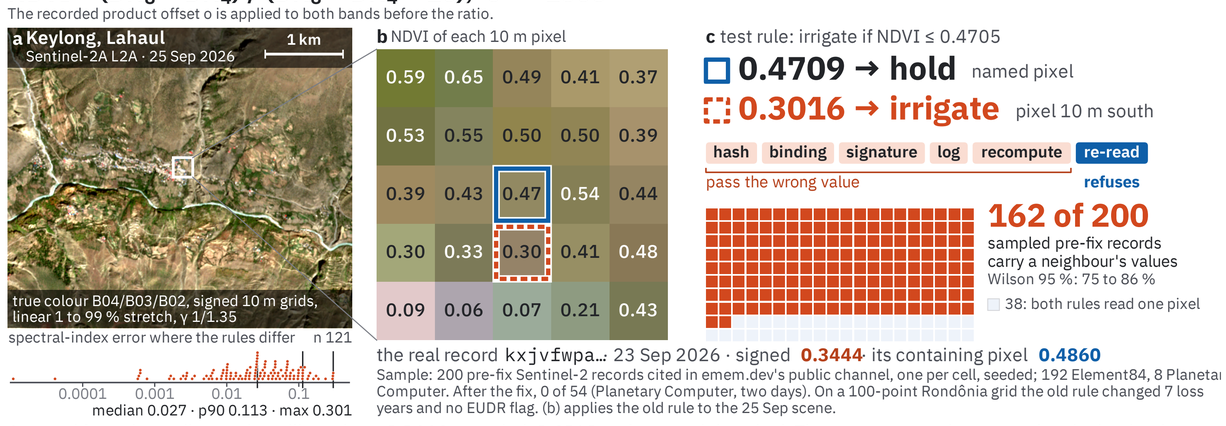
Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 uninstructed, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

6 What does returning to the source add?

A valid record can preserve a wrong source read

$$NDVI = (DN_8 - DN_4) / (DN_8 + DN_4 + 2o),\ o = -1000$$

The recorded product offset o is applied to both bands before the ratio.



the real record `kxjvfwpa…`: 23 Sep 2026 · signed **0.3444** · its containing pixel **0.4860**

Sample: 200 pre-fix Sentinel-2 records cited in emem.dev's public channel, one per cell, seeded; 192 Element84, 8 Planetary Computer. After the fix, 0 of 54 (Planetary Computer, two days). On a 100-point Rondônia grid the old rule changed 7 loss years and no EUDR flag. (b) applies the old rule to the 25 Sep scene.

A record from the earlier reader still resolves: 0.3444 recorded, 0.4860 at the containing pixel. The source re-read exposes the pixel-selection difference.

7 What has the receiver established?

Know what was checked

RECORD	Exact addressed bytes
BINDING	Requested place, time and product
ATTESTATION	Signature under the pinned key
DERIVATION	Declared computation reproduces
SOURCE READ	Named source supports the value

Source quality remains inherited. Entity meaning and downstream decisions belong to the application using the evidence.

8 Can a later agent recover the earlier state?

Memory keeps what was cited

BENGALURU · SAME CELL AND BAND · 2026

same place + band

signed 28 May	signed 11 Aug
918.0 m	915.07 m
Open-Meteo / DEM90	Copernicus DEM30
earlier CID	later CID

as of 15 Jun → 918.0 m

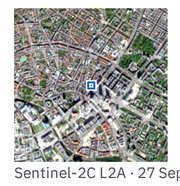
The next agent can recover the earlier citation.

Asked what memory knew on 15 Jun, recall returns 918.0 m. The later 915.07 m record reflects a provider change; it does not establish ground movement.

9 What does one Berlin location reveal?

One place, many observations

Central Berlin: optical, radar, elevation, temperature, water, forest and biomass. One lookup cell organises 15 products, preserving each record and native grid.



BERLIN

`defi.zb655.yaka.pUxe`

52.5243 N, 13.4105 E · one 10 m cell

direct sensor · deterministic index · model output · human curated · unclassified

Sentinel-2C L2A · 27 Sep 2026 · RGB · 2.24 km across; box: 90 m

product	signed reading	valid time	grid	fact_cid
Sentinel-2C L2A	NIR 0.0475	27 Sep 2026	10 m	wtqd64…
Sentinel-2C L2A	NDVI 0.195	27 Sep 2026	10 m	y2k2ti…
Sentinel-1C RTC	VV −2.82 dB	28 Sep 2026	n/s	2x2enx…
GLO-30 DEM	36.6 m DSM	2021	30 m	sdu3gf…
MODIS MOD11A2	297.4 K day LST	29 Aug 2026	1 km	yqhihm…
ESA WorldCover	built-up (50)	2021	10 m	xicg6u…
ESA CCI Biomass	0 t/ha	2022	100 m	keebyc…
JRC GSW v1.4	0 % water	1984 to 2021	30 m	hohgux…
Hansen v1.13	0 % tree cover	2000	n/s	x4kpky…
JRC GFC2020 V4	not forest	2020	10 m	oog2b4…
Copernicus CAMS	NO_2 12.3 µg/m³	30 Sep 2026	~11 km	jvwhwg…
Overture, not EO	0 buildings	Aug 2026	vector	czehax…
ISRIC SoilGrids	*no data, null*	30 Sep 2026	250 m	mkwinv…
CHIRPS v2.0	*outside ±50°*	30 Sep 2026	0.05°	bkaovf…
JRC TMF	*outside ±30°*	30 Sep 2026	30 m	w7t2hx…
NASA FIRMS	*no fire in 24 h*	30 Sep 2026	points	k2ykb2…

15 products, 16 signed facts, 4 signed absences, 10 m to about 11 km; read live 30 Sep 2026. Native grids remain distinct; n/s: grid not in source files. 16 records re-hash under emem.dev's key, 2 Oct 2026.

10 What else can an agent carry?

Beyond a single observation

EARTH DATA
raster → cube → rasterset
Fields and dated stacks; retain the scene.

COLLECT EVIDENCE
8 facts → one bundle handle
38 characters; members resolve separately.

REASONING CHECKPOINTS
located → routed → recalled → scored
4 stage addresses re-hashed offline.

FILES, IDENTITY AND EXECUTION
tree · entity · trace · track
Address chunks, subjects and recorded runs.

Resolve the object and inspect what its address and receipt cover.

NEXT EXPERIMENTS

Measure each cause of change

Validate numeric attribution under controlled scene, pixel and encoder changes. Repeat checked handoffs between separately operated agent hosts.

11 Use emem. Carry the evidence forward.

Connect a plugin, a workflow or your own code. Public emem reads need no API key; the source is open.

Measured separately: the same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).
The routes below are plugins, connectors, packages and listings. Their setup links and workflow examples are behind CONNECT.

CONNECT & REPRODUCE Copy a token, choose your tool, run the checks. `vortx-ai.github.io/esa_poster/use/`

ChatGPT — @emem plugin — Enable emem; paste a token.

Claude — Code plugin · MCP connector — Resolve from a chat or terminal.

Visual Studio Code — MCP gallery: @mcp emem — Install for Copilot agent mode.

Dify — Marketplace plugin — Referent Lock · EUDR workflows

Salesforce MuleSoft — Anypoint Exchange — Vortx AI MCP Server listing

CONNECT
MCP · A2A · REST
Cursor · Gemini CLI

BUILD
Python · TypeScript · self-hosted Docker
LangGraph · LlamaIndex · AutoGen · CrewAI · Mastra

DISCOVER
GitHub MCP · Official MCP Registry
Glama · Smithery · Hugging Face · Context7 · ClawHub

Carry the token → resolve the record → check the evidence → continue in another agent.

Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026. · The linked methods connect this poster's figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)